

Item 5 briefing — Joint retrieval of bottom mixing fractions and depth

Advisor procedure + Item-4 squared Cost | HydroLight validation data | 2026-09-03

Problem

The existing method treated the bottom as a single type and retrieved depth only. Real bottoms are mixtures of several materials, so not only depth but also how much of each material is present (the mixing fractions) must be matched. The comment was not to scan all mixing fractions on a 5–10% grid, but to take the top K by the residual of representative Seeds and then move the mixing fractions and depth together.

Objective

This note records the method that retrieves bottom mixing fractions and depth together. It distinguishes the original method (bottom fixed, depth only) from the method used by the advisor (mixing fractions and depth together). The three-band option was kept as a reference because, in Item 1, the relative error at 665 nm was large; it is not used in the Cost of this calculation (Item 4 showed that dropping 665 nm worsens depth). Results are from 6,250 cases run with the advisor procedure, comparing Cost as-is with Cost multiplied by 1 million, looking at both bottom fractions and depth.

What is used in this calculation

Cost = the sum of squared differences between observed R_{rs} and modelled R_{rs} . This was fixed in Item 4. The advisor's original formula (mean of absolute values) is not used.

The tolerance 0.000001 (10^{-6}) is the SLSQP default. The number itself was not changed. Only when the value is passed to the optimizer is Cost multiplied by 1 million (10^6). The 1 million is the reciprocal of the square of the leftover R_{rs} error (about 0.001) when the true answer is inserted in Item 1.

Validation data

The validation data are mixed-bottom R_{rs} generated with HydroLight. The true bottom to be matched is brown algae 80% + ooid sand 20%. Dark sediment and green algae are 0. The simulation conditions on the presentation slides are given in Table 1. The slides use 11 depth steps and 1,375 cases. This calculation uses the same bottom, water quality and solar zenith angles, but depth only at 0.1 m spacing (50 steps), giving 6,250 cases. The mixed HydroLight set is this single 80:20 grid. Other fractions and pure bottoms are not demonstrated by these numbers.

Table 1. HydroLight validation simulation conditions (presentation slides)

Item	Simulation conditions
Simulation	HydroLight radiative-transfer simulation
Water type	Classic Case-1 (Case-1 Water)
Bottom material (true)	Brown Algae 80% + Ooid Sand 20%
Depth	0.1, 0.5, 1.0, 1.5, ..., 4.5, 5.0 m (11 values) → this calculation: 0.1–5.0 m, 0.1 m spacing (50 values)
Solar zenith angle	10, 20, 30, 40, 50° (5 values)
Wavelength	443, 490, 560, 665 nm (4 values)
Chl-a concentration	0.1, 0.2, 0.3, 0.4, 0.5 mg/m ³ (5 values)
TSM concentration	0.1, 0.2, 0.3, 0.4, 0.5 mg/m ³ (5 values)
Water temperature	20.0 °C
Salinity	35 PSU
Sky / cloud	Clear sky, cloud fraction 0.0
Number of cases	Slides 1,375 cases (11×5×5×5). This calculation 6,250 cases (50×5×5×5)

Procedure

1) Original method — the bottom is already known; only depth is retrieved

The original method assumes the bottom type (reflectance) is known and retrieves depth only. Bottom mixing fractions are not retrieved. Cost is the sum of squared differences between observed R_{rs} and modelled R_{rs} . Because a pixel has one depth, the depth that minimizes this Cost over the four bands 443·490·560·665 nm is found. A depth is not obtained separately in each band and then one of them chosen.

Original method — Cost = sum of squared Rrs errors. Bottom fixed; depth only

$$Cost(z) = \sum_{\lambda} [R_{rs}^{obs}(\lambda) - R_{rs}^{model}(\lambda; z)]^2$$

$$\lambda = 443, 490, 560, 665 \text{ nm}$$

$$z^* = \arg \min_{z > 0} Cost(z)$$

Cost = sum of squared differences between observed Rrs and modelled Rrs

Figure 1. Cost = sum of squared R_{rs} errors. The bottom is fixed, and only the depth z^* that minimizes this value is retrieved. Mixing fractions are not obtained.

Table 2. Symbols in Figure 1

Symbol	Meaning
Cost	Sum of squared differences between observed R_{rs} and modelled R_{rs} . The error in each band is squared and added.
z	Depth to be retrieved (m). One per pixel.
R_{rs}^{obs}	HydroLight (or observed) remote-sensing reflectance
R_{rs}^{model}	Remote-sensing reflectance computed from the shallow-water formula at depth z
λ	Wavelength. The original method uses four: 443, 490, 560, 665 nm
z^*	Depth at which Cost (sum of squared R_{rs} errors) is smallest. Depth is greater than 0.

2) Original method using only three wavelengths

When true depth and true bottom are inserted and R_{rs} is computed, as in Item 1, the relative error at 443·490·560 nm is small, but at 665 nm alone it is 13–25% (Figure 2). R_{rs} itself is small at 665 nm, so the same difference appears large as a relative error. A variant that drops 665 nm from the original Cost and uses only three bands was also kept. In that case too, only depth is retrieved; bottom mixing fractions are not. However, Item 4 showed that dropping 665 nm worsens depth, so the Cost of this calculation keeps four bands. The three-band Cost is a reference and was not used on these 6,250 cases.

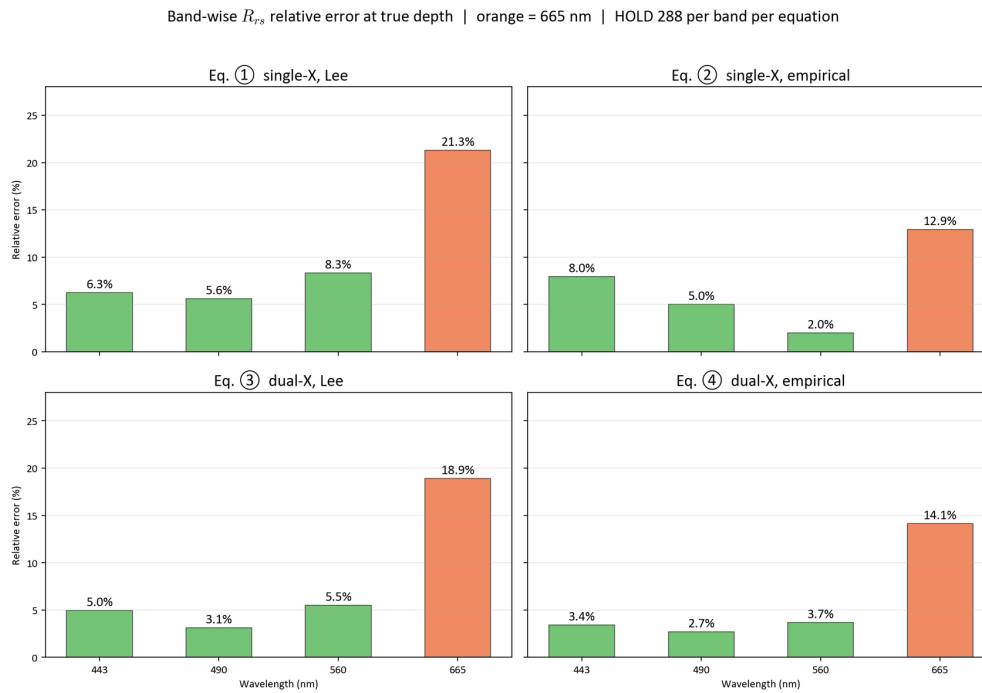


Figure 2. Item 1 validation. Band-wise relative R_{rs} error when the true depth is substituted (HOLD 288). Green = 443-490-560 nm; orange = 665 nm. 665 nm alone is distinctly larger. The three-band Cost is the reference that omits it; the Cost of this calculation keeps four bands.

Three wavelengths only — Cost = sum of squared R_{rs} errors, 665 nm omitted

$$Cost(z) = \sum_{\lambda} [R_{rs}^{obs}(\lambda) - R_{rs}^{model}(\lambda; z)]^2$$

$\lambda = 443, 490, 560 \text{ nm}$ (665 nm excluded)

Figure 3. Three-band Cost = sum of squared R_{rs} errors. The formula is the same as the original method; only the wavelengths are 443-490-560 nm.

Table 3. How Figure 3 differs from the original method

Item	Original method	Three wavelengths only
Wavelengths	443, 490, 560, 665 nm	443, 490, 560 nm
Cost	Sum of squared R_{rs} errors over four bands	Sum of squared R_{rs} errors over three bands
What is retrieved	Depth only (bottom fixed)	Depth only (same)
Mixing fractions	Not retrieved	Not retrieved
This calculation	Cost is this four-band sum of squares	Reference only. Not used on these 6,250 cases

3) Method used by the advisor — bottom mixing fractions and depth are retrieved together

The method used by the advisor does not treat the bottom as a single type. The mixing fractions f_i of the four materials and the depth H are retrieved together. Mixing fractions are not scanned on a full grid. Representative Seeds are made, the mean residual (absolute value) of each Seed is inspected, and only the top K with the smallest residual are taken as initial values; then mixing fractions and depth are moved together. The Cost written by the advisor is not a square; it is the mean of the absolute band errors (J in Figure 4). In this calculation the Seeds are 4 pure + six 50:50 pairs, 10 in total, and the top $K = 5$. Ten is a small number, enough to stand as representatives instead of a full grid, and $K = 5$ is half of that. Three is too few, and ten would leave no selection step.

Three constraints were imposed so that the answer makes sense. ① The four material fractions must sum to 100% (the bottom cannot be empty or 200%). ② A single material cannot be less than 0% or more than 100%. ③ Depth must be greater than 0. It cannot be negative or 0 m. The formula used by the advisor has no upper bound on depth.

This calculation keeps the above procedure (Seed → top K → move together) unchanged, and changes only Cost to the Item-4 sum of squares, and only the depth range to 0.001–6 m. In Item 4 the sum of squares matched depth better, and the depth range is the same as Item 4.

Advisor method — retrieve bottom mixing fractions f and depth H together

1–3. Bottom and initial depth of the representative Seeds

$$\rho_b^{(s)}(\lambda) = \sum_i f_i^{(s)} \rho_i(\lambda), \quad \sum_i f_i^{(s)} = 1$$

$$H_s^{(0)} = \text{DepthEquation}(R_{rs}, \rho_b^{(s)}, a, b_b)$$

4–5. Mean residual of each Seed → top K

$$\varepsilon_s = \frac{1}{M} \sum_{j=1}^M |R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j)|$$

$$S_{TopK} = \text{TopK}_s(\varepsilon_s)$$

6. Optimize mixing fractions and depth together from the top K (SLSQP)

$$\min_{f, H} J(f, H), \quad J = \frac{1}{M} \sum_{j=1}^M |R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j; f, H)|$$

$$\sum_i f_i = 1, \quad 0 \leq f_i \leq 1, \quad H > 0$$

Fractions must sum to 100%; each material 0–100%; depth must be greater than 0.

7–8. Adopt the solution with the smallest final residual among the K

$$\varepsilon_k^{final} = \frac{1}{M} \sum_{j=1}^M |R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j; f_k^*, H_k^*)|$$

$$k^* = \arg \min_k \varepsilon_k^{final}, \quad H^* = H_{k^*}^*, \quad f^* = f_{k^*}^*$$

Figure 4. Procedure used by the advisor. Bottom fractions are set from Seeds (1–3), the top K are selected by mean residual (4–5), mixing fractions and depth are optimized together with SLSQP (6), and the mixing fractions f^* and depth H^* of the solution with the smallest residual are taken (7–8). J in the figure is the advisor's mean of absolute values. The Cost of this calculation is the sum of squares (Table 5).

Table 4. Symbols in Figure 4

Symbol	Meaning
f_i	Mixing fraction of material i. i = ooid, dark sediment, brown algae, green algae
ρ_i	Bottom reflectance of material i
$\rho_b^{(s)}$	Mixed bottom reflectance of Seed s
$H_s^{(0)}$	Initial depth obtained by the original method with that Seed bottom fixed
R_{rs}^{input}	Input (HydroLight) remote-sensing reflectance
R_{rs}^{sim}	Remote-sensing reflectance reconstructed from the formula
ϵ_s, J	Mean of the absolute band errors. M is the number of wavelengths used.
M	Number of wavelengths. The advisor formula is a mean of absolute values. With three bands, M = 3.
K	Number of Seeds passed to continuous optimization. 5 in this calculation.
H^*, f^*	Solution with the smallest final residual among the K. f^* is the bottom mixing fractions; H^* is depth.
$\sum f_i = 1$	The four material fractions must sum to 100%. The bottom cannot be empty or 200%.
$0 \leq f_i \leq 1$	A single material cannot be less than 0% or more than 100%.
$H > 0$	Advisor original formula: depth must be greater than 0, no upper bound. This calculation: 0.001–6 m (Table 5).

Table 5. Settings used in this calculation

Item	Setting
Materials	ooid sand, dark sediment, brown algae, green algae (the four types fixed in Item 3)
True values to match	ooid 20%, brown 80%, dark 0%, green 0%
Seeds	4 pure + six 50:50 pairs, 10 in total. Representatives instead of a full grid.
Top K	The 5 with the smallest initial residual (half of 10. Three is too few; ten would leave no selection step.)
Cost	Sum of squares fixed in Item 4. The advisor's original formula (mean of absolute values) is not used.
Tolerance	0.000001 (10^{-6} , SLSQP default. The number is left as is.)
Value passed to SLSQP	Cost $\times$ 1 million (10^6). Reciprocal of the square of the Item-1 forward R_{rs} error of about 0.001.
Sum of fractions	The four materials must sum to 100%.
Each fraction	Cannot be less than 0% or greater than 100%.
Depth	0.001–6 m (same as Item 4. The advisor's original formula has no upper bound.)

Problem that arose in the procedure

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The stopping criterion (tolerance) was not decided at the outset. 0.000001 (10^{-6}) was used at first because it is the SLSQP default. It was not a number chosen after looking at the size of Cost.

Cost is the sum of squared differences between observed R_{rs} and modelled R_{rs} . R_{rs} itself is small, so Cost is very small, 10^{-7} – 10^{-6} . It is already below the tolerance 0.000001 (10^{-6}), so the optimizer treats it as “already small enough” and stops at once. Depth and bottom mixing fractions are therefore decided quickly, barely moving from the Seed starting values. The mixing fractions are half dark and half brown, and ooid is 0.

The tolerance was not made larger. Making the tolerance larger would stop even more easily. The second calculation left the tolerance number at 0.000001 (10^{-6}) and multiplied only Cost (squared R_{rs} error) by 1 million (10^6) to match the scale of the numbers. The 1 million is the reciprocal of the square of the leftover R_{rs} error (about 0.001) when the true answer is inserted in Item 1. The location of the minimum (argmin) does not change. This is the stopping rule of this calculation.

Results

The tolerance is the SLSQP default 0.000001 (10^{-6}) in both runs. The only difference is whether Cost (sum of squared R_{rs} errors) was passed as-is or multiplied by 1 million (10^6). The bottom to be matched is ooid 20%, brown 80%.

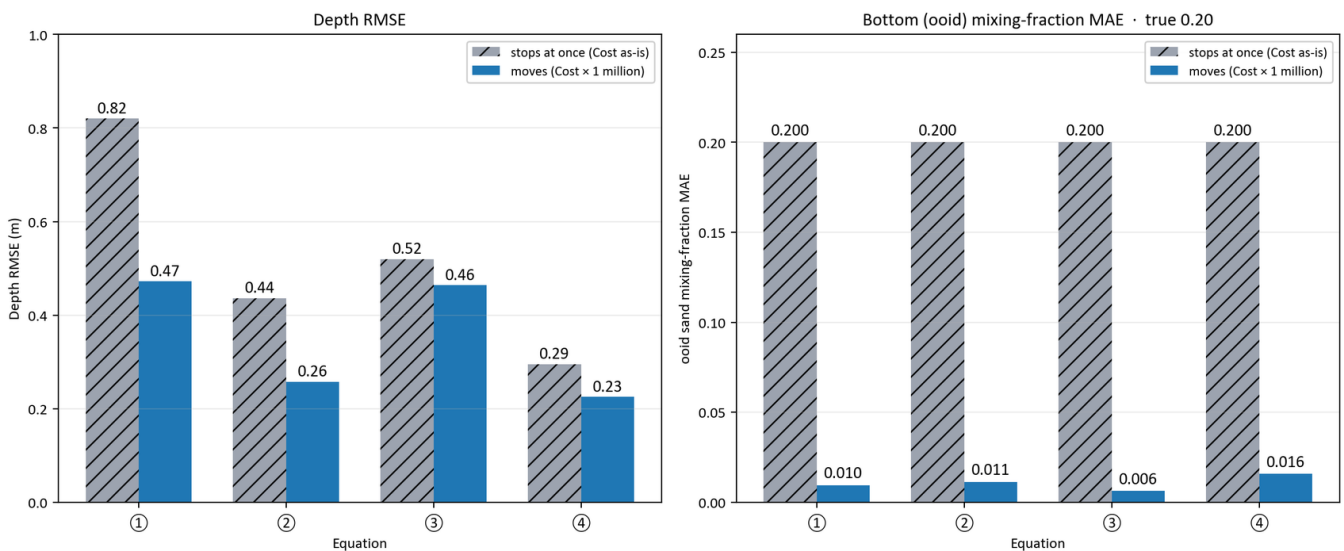


Figure 5. Hatched = Cost as-is (stops at once); blue = Cost × 1 million (moves). Tolerance 0.000001 (10^{-6}) is the SLSQP default. $n = 6250$ per equation.

Looking at Figure 5 alone, Eqs. ② and ④ are good. Depth RMSE is 0.23 m for ④ and 0.26 m for ②. ① and ③ are large, 0.46–0.47 m. The hatched bars on the right are all 0.200 because sand was left at 0. Blue found the sand (0.006–0.016).

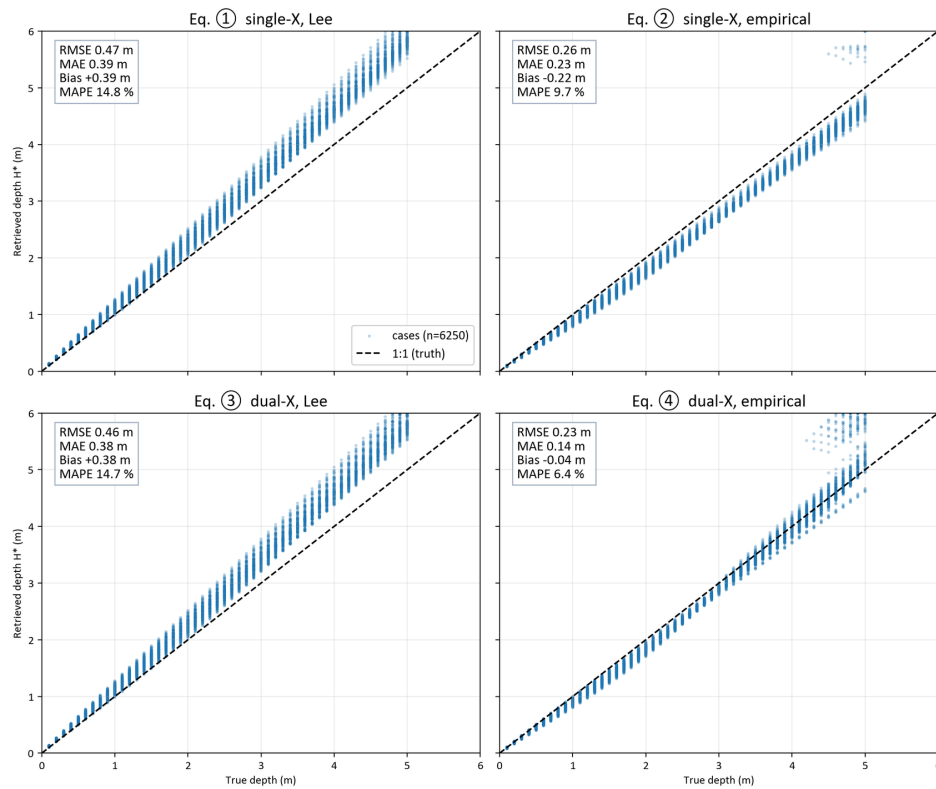
Depth from joint retrieval of mixing fractions and depth | brown 80% + ooid 20% · QAA a , b_b 

Figure 6. Depth scatter. x-axis = true depth; y-axis = retrieved depth. The black dashed line is 1:1 (truth). Closer to the line means depth is matched. $n = 6250$ per equation.

Figure 6 is the figure that shows whether depth is matched, as it is. For Eqs. ② and ④ the points lie near the 1:1 line. Eqs. ① and ③ rise well above the line and take depth too large. This is the same story as the left panel of Figure 5.

Table 6. After Cost $\times$ 1 million — depth, mixing fractions and reconstructed R_{rs} (QAA, $n = 6250$ per equation)

Equation	Depth RMSE (m)	Mean mixing fractions (ooid, dark, brown, green)	ooid MAE	brown MAE	Reconstructed R_{rs} RMSE	R_{rs} relative RMSE (%)
① single-X, Lee coefficients	0.47	(0.19, 0.00, 0.80, 0.01)	0.010	0.012	0.67×10^{-4}	1.11
② single-X, empirical coefficients	0.26	(0.19, 0.00, 0.81, 0.00)	0.011	0.013	0.95×10^{-4}	1.93
③ dual-X, Lee coefficients	0.46	(0.20, 0.01, 0.79, 0.00)	0.006	0.015	0.58×10^{-4}	0.94
④ dual-X, empirical coefficients	0.23	(0.20, 0.02, 0.77, 0.00)	0.016	0.027	1.90×10^{-4}	4.03

Note. True mixing fractions (0.20, 0, 0.80, 0). Blue rows = Eqs. ② and ④, which have good depth in Figure 5. Reconstructed R_{rs} RMSE is the value obtained by running the formula again with the retrieved (mixing fractions, depth) and comparing with HydroLight.

On depth alone, ④ is first. Looking at depth and bottom together, ② is acceptable on both. Reconstructed R_{rs} is smallest for ③ (0.58×10^{-4}) and largest for ④ (1.90×10^{-4}). A good depth does not mean R_{rs} is the best match as well.

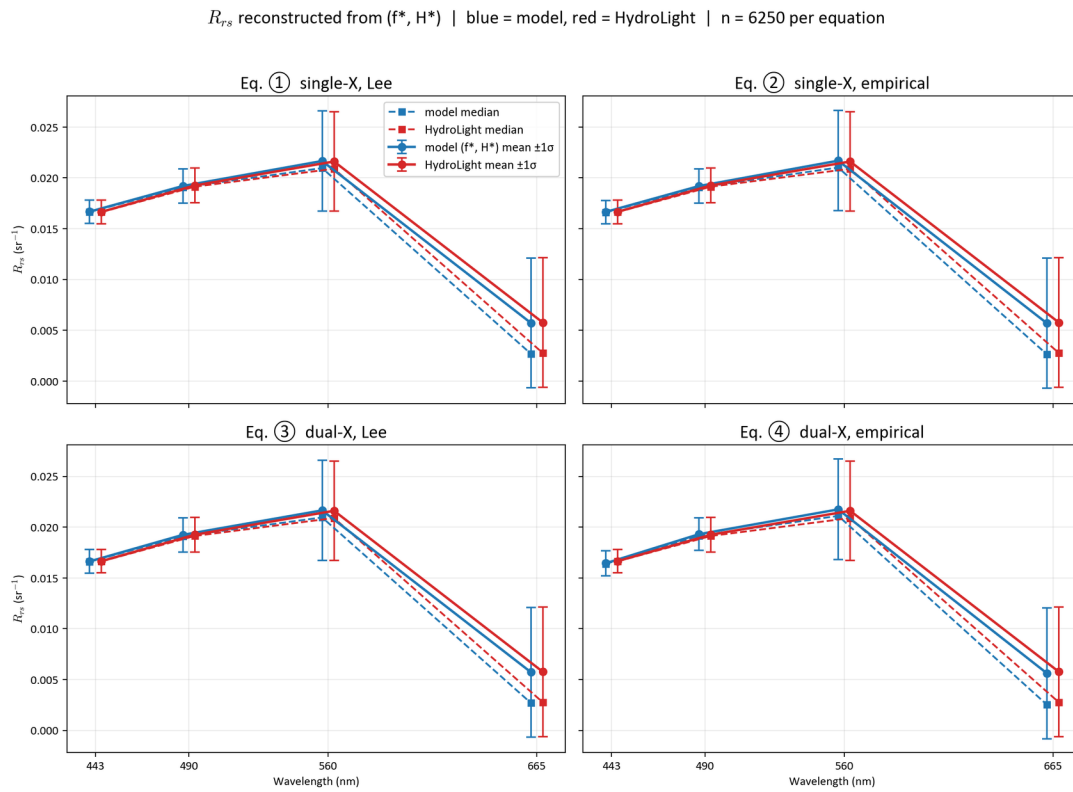


Figure 7. R_{rs} reconstructed from the retrieved mixing fractions and depth (blue) and HydroLight (red). Closer overlap means better reconstruction. $n = 6250$ per equation.

In Figure 7, ① and ③ almost overlap in blue and red. ② is also close. ④ has the best depth, but the spectrum is the most separated. This matches the R_{rs} RMSE in Table 6.

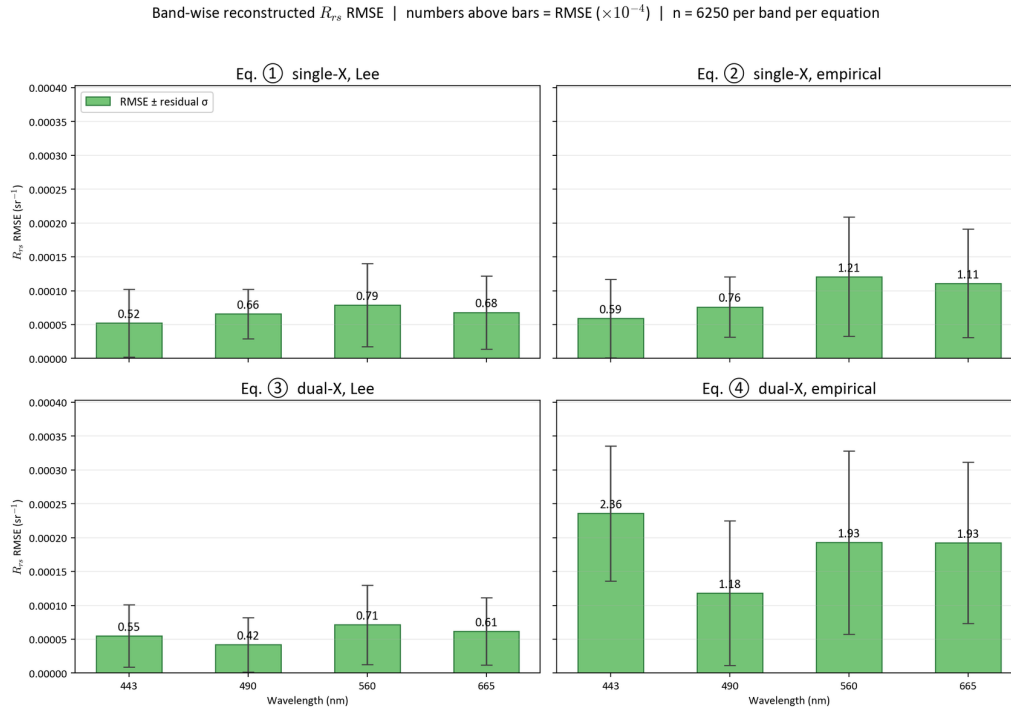


Figure 8. Band-wise reconstructed R_{rs} RMSE. Numbers above the bars are $RMSE \times 10^{-4}$. $n = 6250$ per band per equation.

Figure 8 is the same. The bars for ① and ③ are low, and ④ is high in all four bands. On a mixed bottom, depth and mixing fractions point to ② or ④; once reconstructed R_{rs} is included, ④ is weak.

Remaining limitation

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The stopping rule has been decided. The tolerance number is kept at the SLSQP default 0.000001 (10^{-6}), and Cost is multiplied by 1 million (10^6) to match the scale of the numbers. The 1 million is the reciprocal of the square of the leftover R_{rs} error (about 0.001) when the true answer is inserted in Item 1. The tolerance was not increased.

The remaining limitation is the validation data. The mixed HydroLight set is a single grid of brown algae 80% + ooid sand 20%. On pure bottoms and other mixing fractions these 6,250-case numbers do not demonstrate the result.

Conclusion

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The original method fixes the bottom and retrieves depth only with Cost (the sum of squared differences between observed R_{rs} and modelled R_{rs}). The three-band option was kept as a reference because, in Item 1, the relative error at 665 nm was large; it is not used in this Cost. The method used by the advisor selects the top K by the mean residual of representative Seeds and then jointly optimizes bottom mixing fractions and depth. This calculation keeps that procedure unchanged, and changes only Cost to the Item-4 sum of squares, and only the depth range to 0.001–6 m.

The validation data are a single HydroLight mixed-bottom grid of brown algae 80% + ooid sand 20%. The tolerance 0.000001 (10^{-6}) is the SLSQP default, so it was left as is. Cost is already below the tolerance, so depth and bottom are decided quickly from the starting values; the tolerance is left unchanged and the scale of the numbers is matched by $\text{Cost} \times 1 \text{ million}$ (10^6). The 1 million is the reciprocal of the square of the Item-1 forward R_{rs} error (about 0.001). The location of the minimum does not change. This is what brings the mixing fractions close to the truth. On a mixed bottom, depth is good for ② and ④, and reconstructed R_{rs} is better for ③ and ①. The stopping rule is $\text{Cost} \times 1 \text{ million} + \text{tolerance}$ 0.000001. Other mixing fractions and pure bottoms are left to the next step.